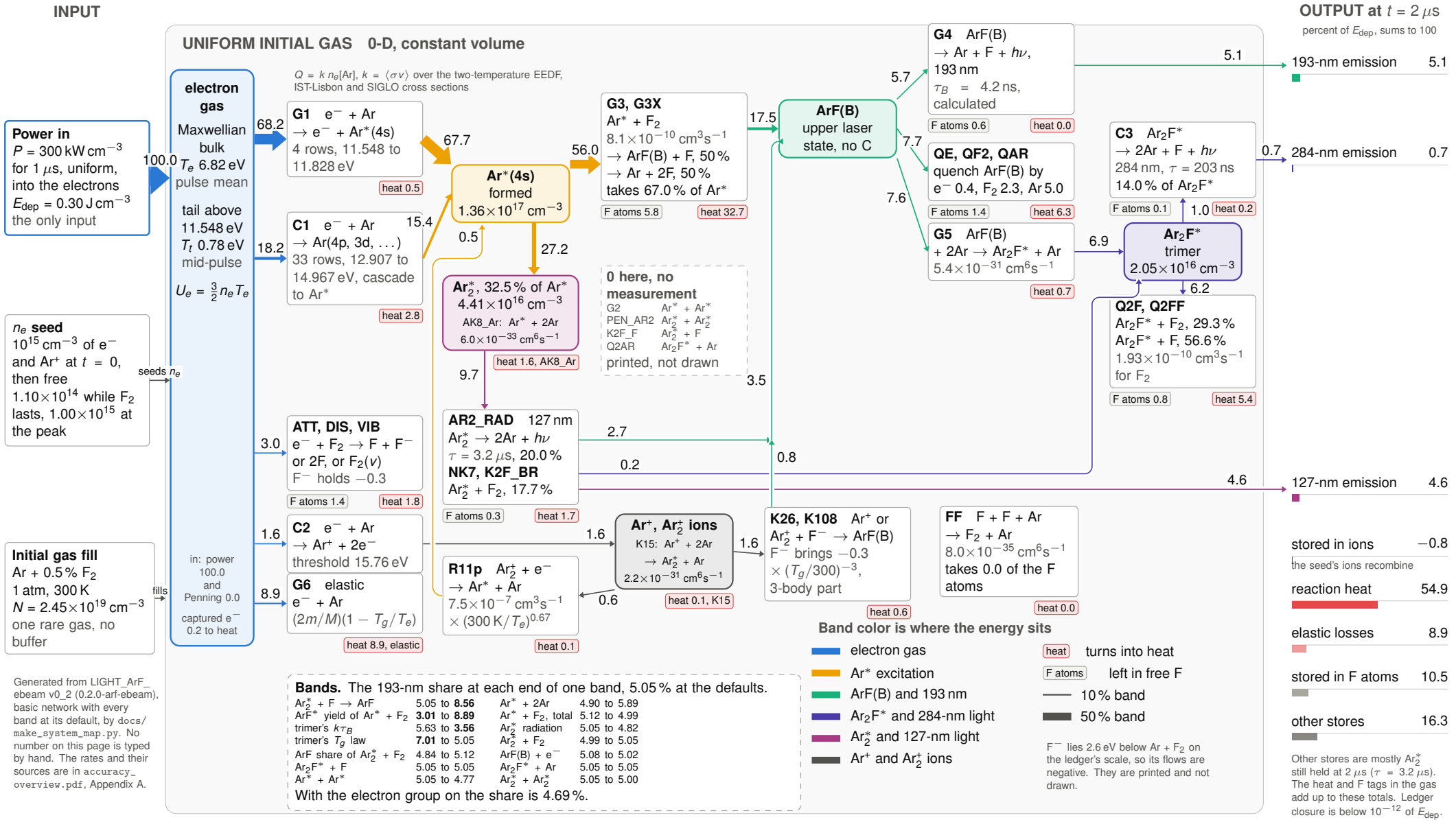
LIGHT_ArF_ebeam system map, v0_2 on LIGHT's own input: energy flow through the Ar/F_2 kinetics

Reference point: 1 atm of Ar with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, every band at its default, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Ar* 86.4 %, to Ar+ 1.6 %, to ArF(B) 18.7 %, heat 54.9 %, elastic losses 8.9 %, emission 10.3 %. Of the Ar* formed, 32.5 % forms Ar_2^* (8.7 % of the Ar* used before the F_2 burns out), and 20.0 % of that Ar_2^* radiates at 127 nm. Of the ions that recombine, 28.6 % meet an electron and return to Ar*, and 71.4 % meet F^- and form ArF(B). Of the ArF(B) formed, 5.46×10^{16} cm^{-3}, 83.2 % came from Ar* + F_2, 3.7 % from ion-ion recombination and 13.1 % from Ar_2^* + F_2. Of it 27.0 % radiates at 193 nm, 36.3 % forms the trimer, and electrons, F_2 and Ar quench 1.9 %, 11.1 % and 23.7 %. At the end of the run T_g = 675 K and p = 2.26 atm, and F_2 fell below a tenth of the fill at 728 ns.